Restoration of *Hydrastis canadensis* by Transplanting with Disturbance Simulation: Results of One Growing Season

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Abstract

Having evolved in an environment with large, severe, and frequent disturbances, including massive floods, fires, and impacts of extinct and extirpated fauna, woodland herbs may be adapted to such disturbance processes. Present lack of such disturbances may contribute to present rarity. We test the hypothesis that transplanting with disturbance simulation can be used to restore the threatened woodland herb, *Hydrastis canadensis* (goldenseal). Three disturbance-simulation treatments (soil turnover, fertilization, and both) and a control were randomly applied to 100 blocks in goldenseal habitat, and a single rhizome was transplanted into each treatment. Transplanting was effective with 85% of the transplants surviving, 41% flowering, and 34% fruiting; thus, transplanting may increase area of occupancy. Soil turnover alone and combined with fertilization caused a significant increase in plant size available to support production of fruit. Increased flower and significantly increased fruit production were also characteristic of soil-turned plots. Results support the hypothesis that some woodland herbs are rare due to lack of certain disturbance, call for consideration of soil disturbance as a potentially important and beneficial influence on woodland herbs regardless of light effects, and suggest that transplanting into soil-overturned plots may restore goldenseal. The assumption that undisturbed conditions are optimal may impede effective management of rare woodland flora, highlighting the need for a more flexible approach.

Key words: ***Hydrastis canadensis*, threatened, risk, restoration, recovery, transplanting, disturbance simulation, soil turnover, fertilization.**

Introduction

The work described here was initiated with the objective of contributing to the restoration of the threatened deciduous woodland herb, *Hydrastis canadensis*, L. (goldenseal). The yellow rhizome of this 1- to 2-foot tall plant with palmate leaves, white flowers, and red berries yields medicinal alkaloids and has become increasingly popular in the general marketplace. Commercial formulations are sold to boost the immune system and to treat common colds, nasal congestion, and infections (Small & Catling 1999; Foster 2000). Virtually all goldenseal on the market is supplied from wild populations in the United States (U.S. Fish and Wildlife Service 1997), and consequently it is perceived to be at a high risk of elimination due to overcollection. It is considered endangered, critically imperiled, imperiled, rare, or uncommon in all 27 states having native populations (U.S. Fish and Wildlife Service 1997). In Canada, goldenseal is considered threatened due to few scattered populations, increasing potential for unrestrained wild harvest, and habitat loss (White 1991; Sinclair & Catling 2000a). Globally, goldenseal was added to Appendix II of CITES (Convention on International Trade in Endangered Species) in 1997 (CITES 2000), which requires artificial propagation for export (Canadian Wildlife Service 1999) and tighter regulation of national and international trade. The factors limiting population increase and the development of methodology for restoration are important considerations for conservation, especially because goldenseal has potential value as both crop and medicine (Small & Catling 1999; Sinclair & Catling 2001).

By restoration, we allude to the need to recover goldenseal from the "threatened" risk category. This can be done by increasing its population size and/or distribution from the category of greater concern (threatened) to one of lesser concern such as "special concern" (see Committee on the Status of Endangered Wildlife in Canada or World Conservation Union categories and criteria, Hilton-Taylor 2000; Catling 2001a), thus reducing the risk of extinction or extirpation.

Recent evidence suggests that naturally occurring populations of goldenseal are not substantially increasing (Sinclair & Catling 2000a, 2002) but potentially benefit from certain kinds of disturbance (Sinclair & Catling 2000a, 2000b). Woodland herbs probably evolved in an environment with large, severe, and frequent soil disturbances that are now no longer prevalent or less severe and frequent. Before extensive settlement by Europeans, disturbances such as severe flooding of bottomland forests, fires,

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and faunal impacts, including those of passenger pigeons (*Ectopistes migratorius* L.), black bears (*Ursus americanus* Pallus), and various very large and numerous Pleistocene mammals (Kurtén & Anderson 1980), may have caused soil disturbance, provided nutrients, and promoted colonization and spread, with or without affecting forest canopies. Catling (2001b) suggested that any eastern North American woodland herb is a candidate for consideration of beneficial effects of a variety of extinct and extirpated animals. We hypothesize that increase in distribution and population size of goldenseal requires certain kinds of natural disturbance. Specifically, in relation to this hypothesis, we address the following questions. Can goldenseal be transplanted successfully? Does soil disturbance and/or soil fertility affect transplant success in terms of recruitment capability?

A large field experiment was used to determine whether transplanting and disturbance simulation, including soil turnover and/or fertilization, within existing habitat benefits goldenseal after a single growing season.

Methods

The Plant

Goldenseal is a perennial herb native to eastern North America, occurring from southern New England to southern Wisconsin and south to Arkansas and northern Georgia (Small & Catling 1999; Sinclair & Catling 2000a). It has a yellow, horizontal, irregularly knotty rhizome with numerous rootlets along its length. One rhizome produces one or more stems per year both from buds on the rhizome and root tips. Stems reach up to 30 cm high and are terminated by one to three palmate leaves, usually with five lobes and as wide as 30 cm. The single white flower, a mass of stamens with no petals, opens in early May (in Ontario) and produces up to 30 seeds enclosed in a green berry that ripens to red in July. The seeds are small, approximately 2 mm, and early seedling development is relatively slow (Small & Catling 1999). Seeds remain dormant in the soil for 10 months and produce a true leaf by the first or second year after germination (personal observation).

Study Sites

The study was carried out at five deciduous woodland sites in the Carolinian zone (Allen et al. 1990) of Ontario, which corresponds to Lake Erie Lowland Ecoregion 135 of the Mixedwood Plains Ecozone (Ecological Stratification Working Group 1995). Four sites occur near the town of Arner (42°02′N, 82°49′W in Gosfield South Township of Essex County) and one near the town of Amherstburg (42°06′N, 83°06′W in Malden Township of Essex County). Exact locations are not given for conservation reasons but are maintained as a confidential file by the Chairman of the Subcommittee for Plants, Committee on the Status of Endangered Wildlife in Canada.

Experimental Design

A random block experiment was set out to test the effects of disturbance on goldenseal transplants. Twenty 3 × 3-m blocks at each of five sites were randomly placed throughout apparent goldenseal habitat in rich deciduous woodland (within 30 m of existing goldenseal patches), permanently marked with aluminum pins and tags, and mapped. The five sites were chosen based on their large population sizes within each site. One day before transplanting, four treatments were randomly applied to each block. Each treatment covered 1 m^2, and treatment plots were 1 m apart to avoid effects of neighboring plots. Treatments were as follows: (1) soil turnover, in which the ground was overturned to 4-cm depth with a shovel; (2) fertilization, in which 7 g of slow release granular bone meal (2:14:0 N:P:K) was evenly sprinkled directly over the ground; (3) both soil turnover and fertilization; and (4) control (no manipulation). The soil disturbance treatment theoretically resembles upheavals created by massive floods or the digging of large extirpated or extinct mammals or erosion after death of flora due to grazing, fertilization, and so on. The fertilization treatment fulfills requirements for field nursery stock based on soil tests (Ontario Ministry of Agriculture, Food, and Rural Affairs 2000) and assumes loss of fertilization events resulting from floods or formerly present fauna including, for example, flocks of passenger pigeons (*Ectopistes migratorius*). Only two levels of nutrients and disturbance were used due to limited time, restricted habitat area, and inadequate population size. It was undesirable to involve more than 10% of the population in experimental manipulation.

From 24 to 27 August 1999, at start of senescence, a single rhizome, from a stem that had flowered or lost fruit, was transplanted with gloves and a trowel into the center of each 1-m^2 treatment plot in the 20 blocks at each of the five sites. The rhizomes were always taken from nearby patches within the site. Stems that had flowered and/or fruited were chosen to eliminate life stage as a factor and to ensure rhizomes of approximately equal size with approximately equal capability to produce photosynthesizing tissue, flowers, and fruit in the following year. Variability in rhizome size and bud number was accounted for by measuring the length and width of rhizomes and counting buds before transplanting.

From 4–6 May, 19–21 June, and 21–23 August 2000, stem height (cm) and leaf width (cm) of transplants were measured. Presence or absence of flowers and fruit were noted. Transplant survival, flowering, and fruiting were calculated as a percentage of the total number of transplanted rhizomes. The amount of photosynthetic tissue was estimated as cover, that is, the maximum surface area of photosynthesizing tissue attained over the growing season. This was calculated as the sum of leaf area (π r^2), stem area (length × width), bud area (π r^2, using an estimated radius of 0.5 cm), and premature fruit area (π r^2, using an average fruit radius of 0.6 cm based on the width of 60 fruit, 3 selected randomly from each of 20 goldenseal populations). Rhizome size was calculated as area (rhizome length × width). Thus, data include initial rhizome size (cm^2) and bud number, total area of photosynthetic surface (cm^2) for each transplant, and number of flowers and fruit in each of four treatments (soil turnover, fertilization, soil turnover + fertilization, control) in each of 20 blocks within each of five sites.

Statistical Analysis

Success of transplanting and relative effectiveness of disturbance simulation with regard to recruitment potential was evaluated through a comparison of percentages of survival, flowering, and fruiting under different treatments. Recruitment potential is defined in terms of increased flowering and fruiting, in conjunction with survival and increase in plant size, and was evaluated with a chi-square test.

The effect of treatment, site, and block-nested-within-site on rhizome mortality, cover, flower production, and fruit production was tested with multifactor analysis of variance (ANOVA). Nesting block within site was appropriate because the experiment involved subgroups (blocks) within samples (sites) and nesting enabled appropriate partitioning of error. The Tukey multiple comparison test was used to distinguish which treatment means were significantly different. The resulting probability values were divided in half to obtain one-tailed values.

Analysis of covariance (ANCOVA) was used to determine whether there was a significant effect of the covariates rhizome size and bud number. Separate linear regressions were used to compare differences between treatments in cover as a result of the covariates.

Results

Eighty-five percent of 400 transplants survived (produced one or more stems), and the percentage survival varied from 74% to 91% over the five sites. Forty-one percent of the 400 transplants produced flowers and 34% produced fruit. Flower production varied from 25% to 60% and fruit production varied from 21% to 45% over the five sites.

ANOVA revealed no significant effect of treatment on rhizome mortality ($p = 0.902$) but a significant effect on cover ($p = 0.011$, $R^2 = 38.9\%$). It was clear that much variation was due to site ($p = 0.000$) and block within site ($p = 0.009$), which was expected on the basis of varying environmental conditions (Table 1). However, there was no interaction between treatment and site. Although the interaction between treatment and block could not be tested because treatments were not replicated within blocks, the fact that blocks explained 40% of the amount of variability in cover compared with treatments (Table 1) suggests that blocks may not have a profound influence on treatments. Based on the Tukey multiple comparison test, the most significant effect was soil turnover and fertilization combined ($p = 0.011$), followed by soil turnover alone ($p = 0.014$), with fertilization insignificant ($p = 0.314$, Fig. 1).

There was no significant effect of disturbance treatments on the number of flowers ($p = 0.128$), but the effect on the number of fruit was almost significant ($p = 0.071$) in the ANOVA (Table 1). The lack of significance appeared to be a consequence of the amount of variation in flower and fruit production as a result of site ($p = 0.000$ and 0.002, respectively, Table 1). In fact, soil turnover resulted in substantially more flowering compared with both the control and other treatments and a significantly higher frequency of fruit production (Table 2). For soil turnover treatments, percentage of fruit produced was double or more at three sites and the same at one site. With respect to the other two treatments, there was a great deal of variation within site, but on average the frequency of fruit production was higher with disturbance simulation (Table 2).

ANCOVA revealed a significant effect of rhizome size ($p = 0.000$) but not bud number ($p = 0.233$) on cover. There was also a significant interaction between treatment and rhizome size ($p = 0.008$). The covariate effect of initial rhizome size was only important with regard to treatments involving fertilization, as illustrated by linear regressions that showed a significant increase in cover with increasing rhizome size for fertilization alone ($p = 0.000$, $R^2 =$

Table 1. ANOVA results of treatment, site, and block-nested-within-site effects on cover, number of flowers, and number of fruit.

Response	*Predictor*	*Sum of Squares*	df	*Mean Square*	*F Ratio*	p
Cover	Treatment	327990.3	3	109330.1	3.77	0.011
	Site	1118161.33	4	279540.33	9.65	0.000
	Block (site)	4025107.56	95	42369.55	1.46	0.009
	Error	8506281.34	297	28977.38		
No. of flowers	Treatment	1.59	3	0.53	1.91	0.128
	Site	6.40	4	1.60	5.78	0.000
	Block (site)	30.79	95	0.32	1.17	0.161
	Error	82.16	297	0.28		
No. fruit	Treatment	1.71	3	0.57	2.36	0.071
	Site	4.13	4	1.03	4.28	0.002
	Block (site)	28.06	95	0.3	1.23	0.102
	Error	71.54	297	0.24		

Sum of squares, degrees of freedom (*df*), mean square, F ratio, and significance (*p*) provided.

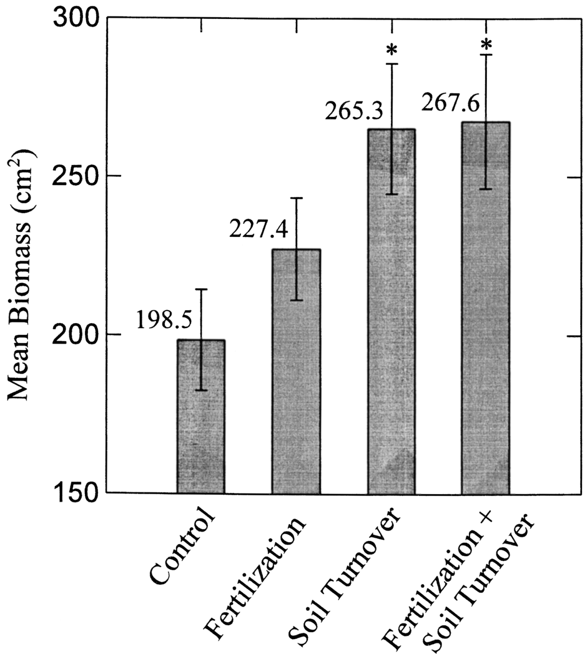

Figure 1. Mean cover (surface area in cm^2) in each treatment after one growing season. Bars represent standard error of the mean. n = 100 for each treatment. *$p < 0.05$.

12.5%) and fertilization combined with soil turnover (p = 0.016, R^2 = 17.3%) but not for soil turnover alone (p = 0.922, R^2 = 0%) or the control (p = 0.322, R^2 = 1.0%).

Discussion

Transplanting Success: Increasing Area of Occupancy

Although transplanting is not to be automatically considered as an appropriate mitigation technique in relation to habitat destruction (Fahselt 1988) and is often unsuccessful in establishing self-sustaining populations (Hall 1987; Fahselt 1988), it is and will continue to be used as a strategy to recover species from "at-risk" categories. In some cases, transplanting may be ineffective due to lack of understanding of existing genetic diversity (Reinartz 1995), natural habitat, or the ecosystem as a whole (Keddy 1983; Fahselt 1988; Lambinon 1993; Schwartz 1994; Morgan 1999). Effectiveness of transplanting ranges from completely ineffective when transplanted propagules such as whole plants or rhizomes die (Hall 1987; Fahselt 1988), to limited effectiveness when seeds of transplants do not germinate (Primack 1996), to very successful when transplants survive and reproduce sexually (Pavlik et al. 1993; Falk et al. 1996). Measures of transplanting success vary and include seedling persistence and maturity, survival of transplanted individuals over the short or long term, and production of fruit and seeds by transplants (Keddy 1983; Primack 1996). Recruitment is considered the highest measure of success (Gordon 1996b; Pavlik 1996; Sutter 1996), because it indicates that the population is self-sustaining through the development of successive generations (Primack 1996). In cases where transplanting resulted in little success, there was limited knowledge of ecological requirements that resulted in poor transplanting technique and management, including planting and timing, site selection and preparation, and postestablishment maintenance (Hall 1987; Falk & Olwell 1992; Guerrant 1996; Primack 1996).

The year 1999 was one of the warmest and driest years on record in the Great Lakes region. The warm temperature and dryness of the soil would likely have retarded development of roots from transplanted rhizomes. In contrast, the year 2000 was cool and wet, which may also have reduced growth due to lack of sunlight and waterlogged soils. Thus, the conditions during the experiment would probably have made successful transplanting more difficult than normal. Eighty-five percent survival of transplants and 41% flowering after transplanting in these difficult years emphasizes the success of transplanting and its potential value for the restoration of goldenseal.

Area of occupancy of goldenseal was increased, and therefore one of the criteria relating to restoration, in terms of transferring a species to a lower risk category, was directly addressed. If there is a sufficient number of plants in the population to allow transplanting and if sites within the apparent habitat can be manipulated to promote spread in the longer term, conservation value of

Table 2. Frequency of stems surviving, flowering, and fruiting within four treatments.

	Survival				*Flowers*				*Fruit*			
Site	*C*	*F*	*T*	*F + T*	*C*	*F*	*T*	*F + T*	*C*	*F*	*T*	*F + T*
1	75	70	80	95	35	35	15	15	30	30	15	15
2	80	70	75	70	30	20	40	30	15	15	30	25
3	85	95	95	80	45	45	45	35	40	30	40	35
4	90	100	85	90	40	55	65	35	30	50	65	30
5	85	90	95	95	40	55	90	55	30	25	75	50
Mean	83	85	86	86	38	42	51	34	29	30	45	31
χ^2		0.04	0.15	0.15		0.19	2.92	0.2		0.00	4.83	0.02
p		0.85	0.7	0.7		0.67	0.09	0.66		1.00	0.03	0.88

Control (C), fertilized (F), soil turnover (T), and fertilized + soil turnover (F+T) at five sites and overall. Yates corrected chi-square (χ^2) is based on comparison of treatment and control means and associated p values are provided. n = 20 for each treatment in each site.

transplanting will be enhanced. This type of "local" transplanting may prove advantageous over augmentation using propagules grown ex situ in terms of adaptation to local habitat conditions, maintenance of the genetic structure, and value of the existing natural habitat. Furthermore, the use of rhizomes, although sometimes unsuccessful with poor site preparation (Hurkmans 1995), may be more effective than attempts at restoration from seed. Experiments involving sowing seeds of rare or extirpated woodland perennial plants have shown that establishment may be difficult, even if the site is experimentally disturbed to enhance the probability of colonization (Primack 1996).

Disturbance Simulation: Increasing Recruitment Potential

Management and monitoring have been emphasized as the most important factors influencing transplanting success (Hall 1987; Bowles et al. 1993; Pavlik et al. 1993; Lusby 1996; McGraw & Levin 1998; Morgan 1999). Management for transplants has included mulching, shade cloth, grazing protection, insecticide, herbicide, and weeding and may be required for years (Guerrant 1996). Management has also involved manipulation of the transplanting site to mimic natural processes of disturbance, including burning and digging, to increase the probability of recruitment (Pullin & Woodell 1987; Watson et al. 1994; Gordon 1996a; Pavlik 1996; Primack 1996). The manipulations (disturbance simulation) are designed to mimic beneficial natural disturbances, the absence of which potentially limits population growth.

Goldenseal clearly benefits from disturbance simulation involving soil turnover through significantly increased cover. There was increased cover in the two treatments that involved soil turnover but not in the treatment with fertilization alone, suggesting that the increase is related to disturbance that causes soil turnover. The relative increased cover produced by initially larger rhizomes in fertilized plots may be a result of a greater capability to take advantage of available nutrients. Although cover was greater under fertilized conditions using large rhizomes, rhizomes of sufficiently large size for a significant increase are not numerous enough to make a difference in a transplanting operation.

In conjunction with this increased cover, more resources were available for reproduction, which was evident in terms of increased fruit production in the soil turnover treatment. The increased opportunity for recruitment through seed by transplants associated with soil turnover addresses the other major criteria for transfer of a species to a lower risk category (i.e., increase in population size).

Broad Implications

Although the extent to which natural disturbance processes were simulated by the field experiment is unknown, the disturbance treatments presumably resemble certain natural disturbances of the past more so than anthropogenic disturbances that exist today, such as trampling, invasive aliens, and harvesting for trees. There may be subtle but significant differences in kinds of disturbance. The results lend support to the concept that to some extent, rare woodland herbs may be rare because of lack of certain kinds of natural disturbance processes to which they are adapted, including severe floods, fires, and faunal impacts (Catling 2001a) that were likely more prevalent in the past.

Goldenseal is an example of a woodland herb that benefits from soil disturbance regardless of light effects. This suggests that canopy gaps are not the only important factors contributing to increase in woodland herb populations despite the extensive literature on the subject (Collins et al. 1985; Moore & Vankat 1986; Collins & Pickett 1988a, 1988b; Jenkins & Parker 1999). The importance of disturbance, other than canopy disturbance, in contributing to increase in a woodland herb has been suggested only recently with very few examples (Thompson 1980; Cook & Lyons 1983; Hughes 1992); most if not all involved canopy alteration.

In summation, transplanting is effective in the restoration of goldenseal, and effectiveness can be increased with soil turnover. The assumption that undisturbed conditions are optimal may impede effective management of rare woodland flora, highlighting the need for a more flexible approach. Experimental transplanting with disturbance simulation may contribute significantly to this flexible approach.

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